

# FDR-Controlled Portfolio Optimization for Sparse Financial Index Tracking

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**Abstract**—In high-dimensional data analysis, such as financial index tracking or biomedical applications, it is crucial to select the few relevant variables while maintaining control over the false discovery rate (FDR). In these applications, strong dependencies often exist among the variables (e.g., stock returns), which can undermine the FDR control property of existing methods like the model-X knockoff method or the T-Rex selector. To address this issue, we have expanded the T-Rex framework to accommodate overlapping groups of highly correlated variables. This is achieved by integrating a nearest neighbors penalization mechanism into the framework, which provably controls the FDR at the user-defined target level. A real-world example of sparse index tracking demonstrates the proposed method's ability to accurately track the S&P 500 index over the past 20 years based on a small number of stocks. An open-source implementation is provided within the R package TRexSelector on CRAN.

**Index Terms**—T-Rex selector, FDR control, high-dimensional variable selection, index tracking, financial engineering.

## I. INTRODUCTION

Financial index tracking is fundamental for the design of asset portfolios that are used to create exchange traded funds (ETFs) and hedging strategies of mutual funds [1], [2]. Prevalent index tracking approaches replicate an entire index (e.g., SPDR S&P 500 ETF) by creating and regularly updating a full tracking portfolio. However, this leads to high transaction costs because it requires the regular purchase and disposition of all assets in an index. Therefore, sparse index tracking methods, which use a small fraction of the stocks that constitute an index, have been proposed [1], [3]–[6]. The common disadvantage of existing sparse approaches is that they require the investor to choose the size of the tracking portfolio or the value of a sparsity tuning parameter. Since there exist no optimal strategies for the choice of these parameters, the authors of the aforementioned approaches resort to experimental choices or rules-of-thumb that often lead to sub-optimal tracking portfolios.

In this work, we propose an approach that automatically determines the size and composition of a sparse index tracking portfolio by controlling the false discovery rate (FDR) [7]. The FDR is the expected fraction of false discoveries (i.e., stocks that are irrelevant for tracking an index) among all selected

stocks. The target FDR between 0 and 100% expresses the level of the investor's willingness to sacrifice a small amount of transaction costs (arising from the inclusion of a few irrelevant stocks into the tracking portfolio) in order to obtain a diversified and yet small tracking portfolio.

Unfortunately, existing FDR-controlling methods are not suitable for sparse financial index tracking, since they are

- 1) not applicable in high-dimensional settings (e.g., [7]–[9]),
- 2) lose the FDR control property in the presence of groups of highly correlated variables (e.g., [10]–[13]), or
- 3) do not account for strongly overlapping groups of highly correlated variables which are characteristic for stock returns data (e.g., [14], [15]).

Therefore, we propose a new FDR-controlling index tracking method by

- 1) extending the dependency-aware *T-Rex* selector [15] to account for strongly overlapping groups of highly correlated variables using a nearest neighbors penalization mechanism,
- 2) proving that it controls the FDR at the investor-specified target level, and
- 3) demonstrating its unique capability of accurately tracking the S&P 500 index over the last 20 years based on a sparse, FDR-controlled, and quarterly updated portfolio.

While this paper highlights an application in finance, the method more generally applies to any setting where the variables exhibit an overlapping groups dependency structure.

**Organization:** Section II briefly revisits the dependency-aware *T-Rex* framework. Sections III and IV introduce the proposed FDR-controlling index tracking method and compare the tracking performance against benchmark methods.

## II. THE DEPENDENCY-AWARE T-REX SELECTOR

The ordinary *T-Rex* selector [11], [12] is a fast FDR-controlling variable selection method for high-dimensional data, where the number of variables  $p$  may exceed the number of samples  $n$ . Its output is an FDR-controlled estimate of the true active variable set  $\mathcal{A} \subseteq \{1, \dots, p\}$  whose cardinality is denoted by  $|\mathcal{A}|$ . The general methodology of the *T-Rex* selector is depicted in Figure 1. First,  $K \cdot L$  dummy predictor vectors of dimension  $n$  are sampled from the univariate standard normal distribution and stacked columnwise into  $K$  different dummy predictor matrices  $\hat{\mathbf{X}}_k \in \mathbb{R}^{n \times L}$ ,  $k = 1, \dots, K$ . These are then appended to the original predictor matrix  $\mathbf{X} \in \mathbb{R}^{n \times p}$ , i.e.,  $\tilde{\mathbf{X}}_k := [\mathbf{X} \ \hat{\mathbf{X}}_k]$ . The enlarged predictor matrices  $\tilde{\mathbf{X}}_k$  and the response vector  $\mathbf{y} \in \mathbb{R}^n$  are the inputs of  $K$  random experiments. In each of the independent experiments, a forward variable selection algorithm, such as the *LARS* [16], *Lasso* [17], or related methods (e.g., [18], [19]), includes one variable in

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The first author has been supported by the LOEWE initiative (Hesse, Germany) within the emergenCITY center. The second author has been supported by the Hong Kong GRF 16206123 research grant. The third author has been supported by the ERC Starting Grant ScReeningData (Project Number: 101042407).

Extensive computations on the Lichtenberg high-performance computer of the Technische Universität Darmstadt were conducted for this research.

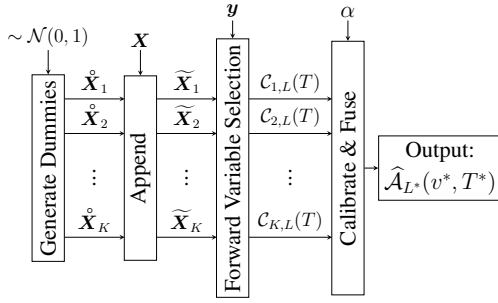


Figure 1: Simplified *T-Rex* selector framework [11], [20].

each forward selection iteration of the algorithm. Each random experiment is early terminated after  $T \in \{1, \dots, L\}$  dummies are included along the forward selection path. The obtained candidate sets  $C_{k,L}(T)$ ,  $k = 1, \dots, K$ , are used to compute the relative occurrences of the variables  $j = 1, \dots, p$ , i.e.,

$$\Phi_{T,L}(j) := \begin{cases} \frac{1}{K} \sum_{k=1}^K \mathbb{1}_k(j, T, L), & T \geq 1 \\ 0, & T = 0 \end{cases}, \quad (1)$$

where the indicator function  $\mathbb{1}_k(j, T, L)$  is equal to one if the  $j$ th variable is contained in the candidate set  $C_{k,L}(T)$  and zero otherwise. Finally, the set of selected variables  $\hat{\mathcal{A}}_L(v, T)$  consists of all variables whose relative occurrences exceed a voting threshold  $v \in [0.5, 1)$ , i.e.,

$$\hat{\mathcal{A}}_L(v, T) := \{j : \Phi_{T,L}(j) > v\}.$$

The extended calibration algorithm proposed in [11] automatically determines a solution  $(v^*, T^*, L^*) \in [0.5, 1) \times \{1, \dots, L^*\} \times \mathbb{N}_+$ , such that the FDR is controlled at the user-specified target level  $\alpha \in [0, 1]$ , while maximizing the true positive rate (TPR). The FDR is the expected value of the false discovery proportion (FDP), i.e.,

$$\text{FDR} := \mathbb{E}[\text{FDP}] := \mathbb{E}\left[\frac{V_{T,L}(v)}{\max\{1, R_{T,L}(v)\}}\right] \leq \alpha.$$

Here,

$$\begin{aligned} V_{T,L}(v) &:= |\{\text{null } j : \Phi_{T,L}(j) > v\}|, \\ R_{T,L}(v) &= |\hat{\mathcal{A}}_L(v, T)| = |\{j : \Phi_{T,L}(j) > v\}| \end{aligned} \quad (2)$$

denote, respectively, the number of selected null variables (i.e., false discoveries) and the total number of selected variables. The TPR is the expected value of the true positive proportion (TPP), i.e.,

$$\text{TPR} := \mathbb{E}[\text{TPP}] := \mathbb{E}\left[\frac{S_{T,L}(v)}{\max\{1, |\mathcal{A}|\}}\right],$$

where  $S_{T,L}(v) := |\{\text{active } j : \Phi_{T,L}(j) > v\}|$  denotes the number of selected true active variables.

The dependency-aware *T-Rex* (*T-Rex+DA*) selector [15] is an extension of the *T-Rex* selector that accounts for dependencies among variables by replacing the ordinary relative occurrences in (1) by dependency-aware relative occurrences

$$\Phi_{T,L}^{\text{DA}}(j, \rho_{\text{thr}}) := \Psi_{T,L}(j, \rho_{\text{thr}}) \cdot \Phi_{T,L}(j),$$

where the penalty factor  $\Psi_{T,L}(j, \rho_{\text{thr}}) \in [0.5, 1]$  is defined by

$$\Psi_{T,L}(j, \rho_{\text{thr}}) := \begin{cases} 1 \\ 2 - \min_{j' \in \text{Gr}(j, \rho_{\text{thr}})} \{|\Phi_{T,L}(j) - \Phi_{T,L}(j')|\} \\ 1/2, \end{cases} \quad \begin{aligned} &\text{Gr}(j, \rho_{\text{thr}}) \neq \emptyset \\ &\text{Gr}(j, \rho_{\text{thr}}) = \emptyset \end{aligned} \quad (3)$$

$\Psi_{T,L}(j, \rho_{\text{thr}})$  penalizes the  $j$ th ordinary relative occurrence according to its resemblance with other ordinary relative occurrences in its associated variable group  $\text{Gr}(j, \rho_{\text{thr}}) \subseteq \{1, \dots, p\}$ . The parameter  $\rho_{\text{thr}}$  controls the size of the variable groups and is automatically determined by the *T-Rex+DA* calibration algorithm such that the FDR is controlled. In [15], it has been proposed to use a hierarchical clustering approach to design disjoint variable groups that allow for FDR control in the presence of disjoint groups of highly correlated variables.

### III. PROPOSED: FDR-CONTROLLED INDEX TRACKING

The existing design of  $\text{Gr}(j, \rho_{\text{thr}})$  for the dependency-aware *T-Rex* framework [15] does not allow for arbitrary overlapping groups of highly correlated variables, which are characteristic for stock returns data. Therefore, in this section, we propose a new design for the variable groups  $\text{Gr}(j, \rho_{\text{thr}})$ ,  $j = 1, \dots, p$ , which incorporates a nearest neighbors (NN) penalization mechanism into the dependency-aware *T-Rex* framework. We then prove that the proposed approach controls the FDR at the user-specified target level. Finally, the stock returns data model is introduced and the proposed FDR-controlled index tracking algorithm is formulated.

#### A. Group Design and FDR Control

For the proposed NN group design, each variable  $j$  is assigned a variable group that contains variables whose correlations with variable  $j$  exceed a threshold  $\rho_{\text{thr}} \in [0, 1]$ , i.e.,

$$\text{Gr}(j, \rho_{\text{thr}}) := \{j' \in \{1, \dots, p\} \setminus \{j\} : |\text{corr}(\mathbf{x}_j, \mathbf{x}_{j'})| \geq \rho_{\text{thr}}\}. \quad (4)$$

Clearly, in contrast to [15], this group design allows for overlapping groups of highly correlated variables. The challenge now is to jointly determine  $\rho_{\text{thr}}$  in (4) and the other parameters of the dependency-aware *T-Rex* selector (i.e.,  $v$ ,  $T$ , and  $L$ ) such that the proposed NN group design yields FDR-controlled solutions

$$\hat{\mathcal{A}}_L(v, \rho_{\text{thr}}, T) := \{j : \Phi_{T,L}^{\text{NN}}(j, \rho_{\text{thr}}) > v\},$$

where

$$\Phi_{T,L}^{\text{NN}}(j, \rho_{\text{thr}}) := \Psi_{T,L}^{\text{NN}}(j, \rho_{\text{thr}}) \cdot \Phi_{T,L}(j) \quad (5)$$

is the dependency-aware relative occurrence of the  $j$ th variable using the proposed NN group design in (4).

Note that the distinction of the penalty functions  $\Psi_{T,L}$  and  $\Psi_{T,L}^{\text{NN}}$  in (3) and (5) stems from their underlying group designs. The underlying group design of  $\Psi_{T,L}$  only allows for disjoint variable groups. In contrast, the proposed NN group design in (4) extends the scope of the *T-Rex+DA* selector by allowing for overlapping groups of highly correlated variables, which are characteristic for applications such as the considered financial index tracking.

In order to prove that the proposed NN group design controls the FDR, we first define the conservative FDP estimator<sup>1</sup>

$$\widehat{\text{FDP}}(v, \rho_{\text{thr}}, T, L) := \frac{\widehat{V}_{T,L}(v, \rho_{\text{thr}})}{\max\{1, R_{T,L}(v, \rho_{\text{thr}})\}}, \quad (6)$$

where  $R_{T,L}(v, \rho_{\text{thr}}) := |\widehat{\mathcal{A}}(v, \rho_{\text{thr}}, T)|$  is the number of selected variables. Second, let  $\widehat{\mathcal{A}}(v, \rho_{\text{thr}}, T) := \widehat{\mathcal{A}}_L(v, \rho_{\text{thr}}, T)$  and  $\Delta\Phi_{t,L}^{\text{NN}}(j, \rho_{\text{thr}}) := \Phi_{t,L}^{\text{NN}}(j, \rho_{\text{thr}}) - \Phi_{t-1,L}^{\text{NN}}(j, \rho_{\text{thr}})$ . Then, the proposed estimator of the unknown number of selected null variables  $V_{T,L}(v, \rho_{\text{thr}})$  in (6) is defined by

$$\begin{aligned} \widehat{V}_{T,L}(v, \rho_{\text{thr}}) &:= \sum_{j \in \widehat{\mathcal{A}}(v, \rho_{\text{thr}})} \left(1 - \Phi_{T,L}^{\text{NN}}(j, \rho_{\text{thr}})\right) \\ &+ \underbrace{\sum_{t=1}^T \frac{p - \sum_{q=1}^p \Phi_{t,L}^{\text{NN}}(q, \rho_{\text{thr}})}{L - (t-1)} \cdot \frac{\sum_{j \in \widehat{\mathcal{A}}(v, \rho_{\text{thr}})} \Delta\Phi_{t,L}^{\text{NN}}(j, \rho_{\text{thr}})}{\sum_{j \in \widehat{\mathcal{A}}(0.5, \rho_{\text{thr}})} \Delta\Phi_{t,L}^{\text{NN}}(j, \rho_{\text{thr}})}}_{=: \widehat{V}'_{T,L}(v, \rho_{\text{thr}})}, \end{aligned} \quad (7)$$

which is similar to the estimator that was defined in [11], [15] with the innovation that the ordinary relative occurrences are replaced by the proposed relative occurrences in (5).

With all necessary definitions in place, we now state and prove that any quadruple  $(v, \rho_{\text{thr}}, T, L) \in [0.5, 1) \times [0, 1] \times \{1, \dots, L\} \times \mathbb{N}_+$ , for which the conservative FDP estimator in (6) does not exceed the user-specified target level  $\alpha \in [0, 1]$ , yields FDR control. To simultaneously maximize the number of selected variables (to obtain highest possible TPR while controlling FDR), we choose the lowest possible voting threshold  $v$  that maintains FDR control, i.e.,

$$v := \inf\{\nu \in [0.5, 1) : \widehat{\text{FDP}}(\nu, \rho_{\text{thr}}, T, L) \leq \alpha\}. \quad (8)$$

**Theorem 1 (FDR control).** *Let  $\text{Gr}(j, \rho_{\text{thr}})$  be as defined in (4) and  $K \rightarrow \infty$ . Suppose that  $\widehat{V}'_{T,L}(v, \rho_{\text{thr}}) > 0$ . Then, for any quadruple  $(v, \rho_{\text{thr}}, T, L) \in [0.5, 1) \times [0, 1] \times \{1, \dots, L\} \times \mathbb{N}_+$  that satisfies Eq. (8), it holds that  $\text{FDR}(v, \rho_{\text{thr}}, T, L) \leq \alpha$ , i.e., FDR control at the user-specified target level  $\alpha \in [0, 1]$ .*

*Proof.* An upper bound on the FDR is given by

$$\begin{aligned} \text{FDP}(v, \rho_{\text{thr}}, T, L) &= \frac{V_{T,L}(v, \rho_{\text{thr}})}{\max\{1, R_{T,L}(v, \rho_{\text{thr}})\}} \\ &= \widehat{\text{FDP}}(v, \rho_{\text{thr}}, T, L) \cdot \frac{V_{T,L}(v, \rho_{\text{thr}})}{\widehat{V}_{T,L}(v, \rho_{\text{thr}})} \\ &\leq \alpha \cdot \frac{V_{T,L}(v, \rho_{\text{thr}})}{\widehat{V}_{T,L}(v, \rho_{\text{thr}})}, \\ &\leq \alpha \cdot \frac{V_{T,L}(v, \rho_{\text{thr}})}{\widehat{V}'_{T,L}(v, \rho_{\text{thr}})}, \end{aligned}$$

where the second, third, and fourth line follow from (6), (8), and (7), respectively. Taking the expected value yields

$$\text{FDR}(v, \rho_{\text{thr}}, T, L) \leq \alpha \cdot \mathbb{E} \left[ \frac{V_{T,L}(v, \rho_{\text{thr}})}{\widehat{V}'_{T,L}(v, \rho_{\text{thr}})} \right].$$

<sup>1</sup>Conservative is meant in the sense that  $\mathbb{E}[\text{FDP}] \leq \mathbb{E}[\widehat{\text{FDP}}]$  [11].

It remains to prove that  $\mathbb{E}[V_{T,L}(v, \rho_{\text{thr}})/\widehat{V}'_{T,L}(v, \rho_{\text{thr}})] \leq 1$ . Along the lines of the proof of Lemma 1 in [15], it can be shown that  $V_{T,L}(v, \rho_{\text{thr}})/\widehat{V}'_{T,L}(v, \rho_{\text{thr}})$  is a backward-running super-martingale with respect to the filtration  $\mathcal{F}_v := \sigma(\{V_{T,L}(v, \rho_{\text{thr}})\}_{u \geq v}, \{\widehat{V}'_{T,L}(v, \rho_{\text{thr}})\}_{u \geq v})$ . Therefore, and since  $v$  in (8) is  $\mathcal{F}_v$ -measurable, an upper bound on  $\mathbb{E}[V_{T,L}(v, \rho_{\text{thr}})/\widehat{V}'_{T,L}(v, \rho_{\text{thr}})]$  can be derived using Doob's optional stopping theorem [21]. First, using the definition of  $V_{T,L}(v)$  in (2) and defining  $\widehat{V}'_{T,L}(0.5) := \sum_{t=1}^T \frac{p - \sum_{q=1}^p \Phi_{t,L}^{\text{NN}}(q)}{L - (t-1)}$ , we obtain

$$\mathbb{E} \left[ \frac{V_{T,L}(v, \rho_{\text{thr}})}{\widehat{V}'_{T,L}(v, \rho_{\text{thr}})} \right] \leq \mathbb{E} \left[ \frac{V_{T,L}(0.5, \rho_{\text{thr}})}{\widehat{V}'_{T,L}(0.5, \rho_{\text{thr}})} \right] \quad (9)$$

$$\leq \mathbb{E} \left[ \frac{V_{T,L}(0.5)}{\widehat{V}'_{T,L}(0.5)} \right] \quad (10)$$

$$\leq 1. \quad (11)$$

Inequality (9) follows from Doob's optional stopping theorem. Inequality (10) follows from the following two inequalities:

$$\begin{aligned} \text{(i)} \quad V_{T,L}(0.5, \rho_{\text{thr}}) &= |\{\text{null } j : \Phi_{T,L}^{\text{NN}}(j, \rho_{\text{thr}}) > 0.5\}| \\ &\leq |\{\text{null } j : \Phi_{T,L}(j) > 0.5\}| = V_{T,L}(0.5) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \widehat{V}'_{T,L}(0.5, \rho_{\text{thr}}) &= \sum_{t=1}^T \frac{p - \sum_{q=1}^p \Phi_{t,L}^{\text{NN}}(q, \rho_{\text{thr}})}{L - (t-1)} \\ &\geq \sum_{t=1}^T \frac{p - \sum_{q=1}^p \Phi_{t,L}(q)}{L - (t-1)} = \widehat{V}'_{T,L}(0.5). \end{aligned}$$

where the inequalities in (i) and (ii) both follow from Eq. (5) and the fact that  $\Psi_{T,L}^{\text{NN}}(j, \rho_{\text{thr}}) \leq 1$ . Finally, inequality (11) has already been proven to hold (see proof of Theorem 1 in [11]) and, thus, it holds that  $\text{FDR}(v, \rho_{\text{thr}}, T, L) \leq \alpha$ .  $\square$

## B. Stock Returns Data Model

As suggested in [1], we model the stocks returns as a linear regression model  $\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}$ , where  $\mathbf{w} = [w_1 \dots w_p]^\top \in \mathbb{R}^p$  is the asset weight vector and  $\boldsymbol{\epsilon} = [\epsilon_1 \dots \epsilon_n]^\top \in \mathbb{R}^n$  is an additive noise vector. Here,  $\mathbf{y} = [y_1 \dots y_n]^\top \in \mathbb{R}^n$  is the daily index returns vector, i.e.,  $y_i = (\text{index}_i - \text{index}_{i-1}) / \text{index}_{i-1}$ ,  $i = 1, \dots, n$ , where  $\text{index}_i$  is the closing price of the index on day  $i$  and  $\text{index}_0$  is the closing price at the first day of the considered period. Analogously,  $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_p] \in \mathbb{R}^{n \times p}$  is the matrix containing the daily returns of the stocks  $\mathbf{x}_j = [x_{1,j} \dots x_{n,j}]^\top \in \mathbb{R}^n$ , i.e.,  $x_{i,j} = (\text{price}_{i,j} - \text{price}_{i-1,j}) / \text{price}_{i-1,j}$ ,  $i = 1, \dots, n$ , where  $\text{price}_{i,j}$  is the closing price of the  $j$ th stock on day  $i$  and  $\text{price}_{0,j}$  is the closing price on the first day of the period.

## C. Algorithm: FDR-Controlled Index Tracking

The general goal of sparse index tracking is to determine a sparse estimator of the asset weight vector  $\mathbf{w}$  that tracks the index  $\mathbf{y}$  sufficiently well using few relevant assets while obeying the following two rules [1]:

1. Shorting stocks is not allowed, i.e.,  $w_j \geq 0$ ,  $j = 1, \dots, p$ .
2. The available budget has to be invested, i.e.,  $\|\mathbf{w}\|_1 = 1$ .

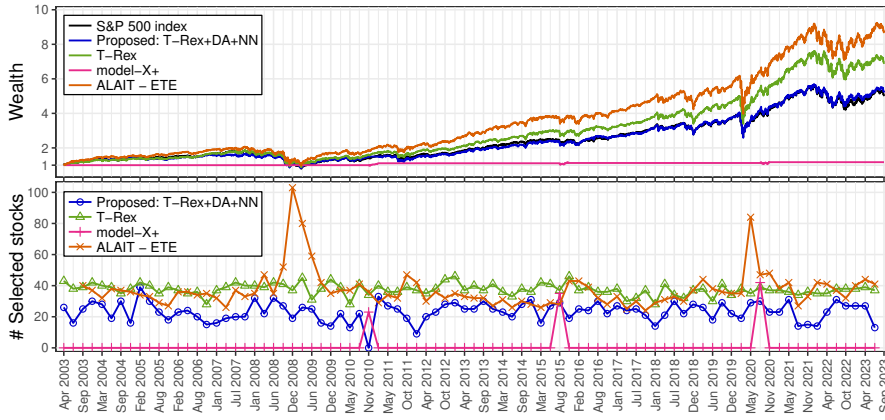


Figure 2: The proposed *T-Rex+DA+NN* selector closely follows the S&P 500 index using the fewest number of stocks in almost all quarters.

The proposed algorithm regularly updates the FDR-controlled tracking portfolio in a rolling-window fashion, as suggested in [1], while satisfying the “no-shorting” and budget constraint in every training period  $m \in \{1, \dots, M\}$ . Therefore, in every training period  $m$ , we first run the dependency-aware *T-Rex* selector with the proposed nearest neighbors group design in (4) and then solve the constrained quadratic problem

$$\begin{aligned} \min_{\mathbf{w}_m} & \|\mathbf{y}_m - \mathbf{X}_{m, \hat{\mathcal{A}}^{(m)}} \cdot \mathbf{w}_m\|_2^2 + \lambda_2 \|\mathbf{w}_m\|_2^2 \\ \text{subject to} & \quad \mathbf{w}_m \geq \mathbf{0} \text{ and } \|\mathbf{w}_m\|_1 = 1. \end{aligned} \quad (12)$$

Here,  $\mathbf{X}_{m, \hat{\mathcal{A}}^{(m)}}$  contains only the daily returns of the selected subset of stocks in  $\hat{\mathcal{A}}^{(m)}$  and the penalty term  $\lambda_2 \|\mathbf{w}_m\|_2^2$  with  $\lambda_2 = 10^{-9}$  ensures a unique solution when the number of selected stocks exceeds the number of trading days in a training period. Algorithm 1 summarizes the proposed index tracking method. An open-source implementation of the proposed and related methods [22], [23] is available within the R package ‘TRexSelector’ on CRAN [24].

#### IV. TRACKING THE S&P 500 INDEX

We consider the S&P 500 index in the 20 year period from 01/01/2003–29/09/2023, which gives us 5,220 trading days (i.e., samples) in total. These are divided into  $M = 86$  training and 86 testing periods, where the first period can only be used for training and the last period only for testing. Using the same rolling window approach as in [1], the portfolio is updated quarterly, i.e., all training and testing periods consist of  $n = 60$  trading days. After removing all stocks that contain missing values,  $p = 390$  candidate stocks are left.

As FDR-controlling benchmark methods, we consider the ordinary *T-Rex* selector [11] and the *model-X* knockoff+ method [10] for the variable selection step in Algorithm 1. We also consider the non-FDR-controlling state-of-the-art sparse index tracking method *ALAIT - ETE* [1] that solves a Lasso-type optimization problem with the same no-shorting and budget constraints using a majorization minimization approach. The target FDR  $\alpha$  is set to 30% and the sparsity tuning parameter for *ALAIT - ETE* to  $\lambda = 10^{-7}$ , as suggested in [1], [25]. The tracking performance is measured by the more comprehensible wealth. That is, we do not consider the absolute value of the index but set the value of the index

|           |          | <i>ALAIT - ETE</i> |      | <i>T-Rex+DA+NN</i> |             |
|-----------|----------|--------------------|------|--------------------|-------------|
| $\lambda$ | $\alpha$ | # stocks           | MSTE | # stocks           | MSTE        |
| 16.67     | 20       | 12.72              | 0.96 | <b>9.62</b>        | <b>0.91</b> |
| 6.67      | 25       | 18.73              | 0.76 | <b>13.67</b>       | <b>0.03</b> |
| 3.33      | 30       | 24.23              | 1.32 | <b>19.72</b>       | <b>0.22</b> |

Table I: The proposed *T-Rex+DA+NN* selector achieves a lower average (over all quarters) mean squared wealth tracking error (MSTE) using a smaller portfolio. Another major advantage is that the proposed method only requires choosing an interpretable target FDR level  $\alpha$  (in %), while the benchmark method relies on the uninterpretable tuning parameter  $\lambda$  ( $\times 10^{-7}$ ).

#### Algorithm 1 FDR-controlled index tracking.

1. **Input:**  $\alpha \in [0, 1]$ ,  $\mathbf{X}$ ,  $\mathbf{y}$ .
2. **For**  $m = 1, \dots, M$  **do**:
  - 2.1. **Compute**  $\text{Gr}(j, \rho_{\text{thr}})$ ,  $j = 1, \dots, p$ , as defined in (4) for all  $\rho_{\text{thr}} \in \{0, 0.01, 0.02, \dots, 1\}$ .
  - 2.2. **Run** the dependency-aware *T-Rex* selector (Algorithm 1 in [15]) with the proposed groups  $\text{Gr}(j, \rho_{\text{thr}})$  in (4) to obtain an optimal solution  $(v^*, \rho_{\text{thr}}^*, T^*, L^*)$  and the corresponding FDR-controlled set of selected stocks  $\hat{\mathcal{A}}^{(m)} := \hat{\mathcal{A}}_{L^*}^{(m)}(v^*, \rho_{\text{thr}}^*, T^*, L^*)$ .
  - 2.3. **Solve** the quadratic optimization problem in (12).
3. **Output:** Portfolio weight vectors for each quarter, i.e.,  $\hat{\mathbf{w}}_1, \dots, \hat{\mathbf{w}}_M$ .

at the start of the tracking period to one and the wealth represents how an investors wealth has changed with respect to the reference. A mathematical definition of the wealth is given in [1]. Note that index tracking does not aim at achieving the highest possible wealth but at an accurate tracking of the wealth corresponding to the index.

The results in Figure 2 and Table I show that the proposed dependency-aware *T-Rex* selector with the nearest neighbors penalization mechanism (*T-Rex+DA+NN*) has the best index tracking performance, while requiring the smallest number of stocks in almost all quarters. The *model-X* knockoff+ method selects no stocks in almost all quarters and, therefore, does not track the index.

#### V. CONCLUSION

The dependency-aware *T-Rex* framework has been extended to account for overlapping groups of highly correlated variables, which are characteristic for stock returns data. The proposed method provably controls the FDR and has been successfully applied in tracking the S&P 500 index over the last 20 years with higher accuracy and fewer stocks compared to state-of-the-art methods. The proposed method does not require sparsity parameter tuning. Therefore, it is a promising approach for index tracking as well as other applications where overlapping groups of highly correlated variables exist.

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